

PROTEX: HACK-2-WIN
(PROTE`G`E, IGD TUW)

TrackMyGas

“Making LPG distribution fair, transparent, and intelligent for every household”

TEAM: LPG LEGENDS

TRACK: OPEN INNOVATION

“TRANSPARENCY BUILDS TRUST.”



PROBLEM STATEMENT

In recent times, global geopolitical tensions and conflicts have disrupted energy supply chains.

At the ground level, the situation worsens due to **system inefficiencies and exploitation**:

- Cylinders are **diverted and sold illegally at higher prices**.
- Some distributors **hoard stock** to create artificial scarcity
- Multiple bookings are made using different identities, blocking genuine users
- Consumers are forced to pay **above government-fixed prices** just to get timely delivery.

“Lack of transparency, coordination, and control in LPG distribution leads to shortages, black marketing, and unfair access for consumers.”

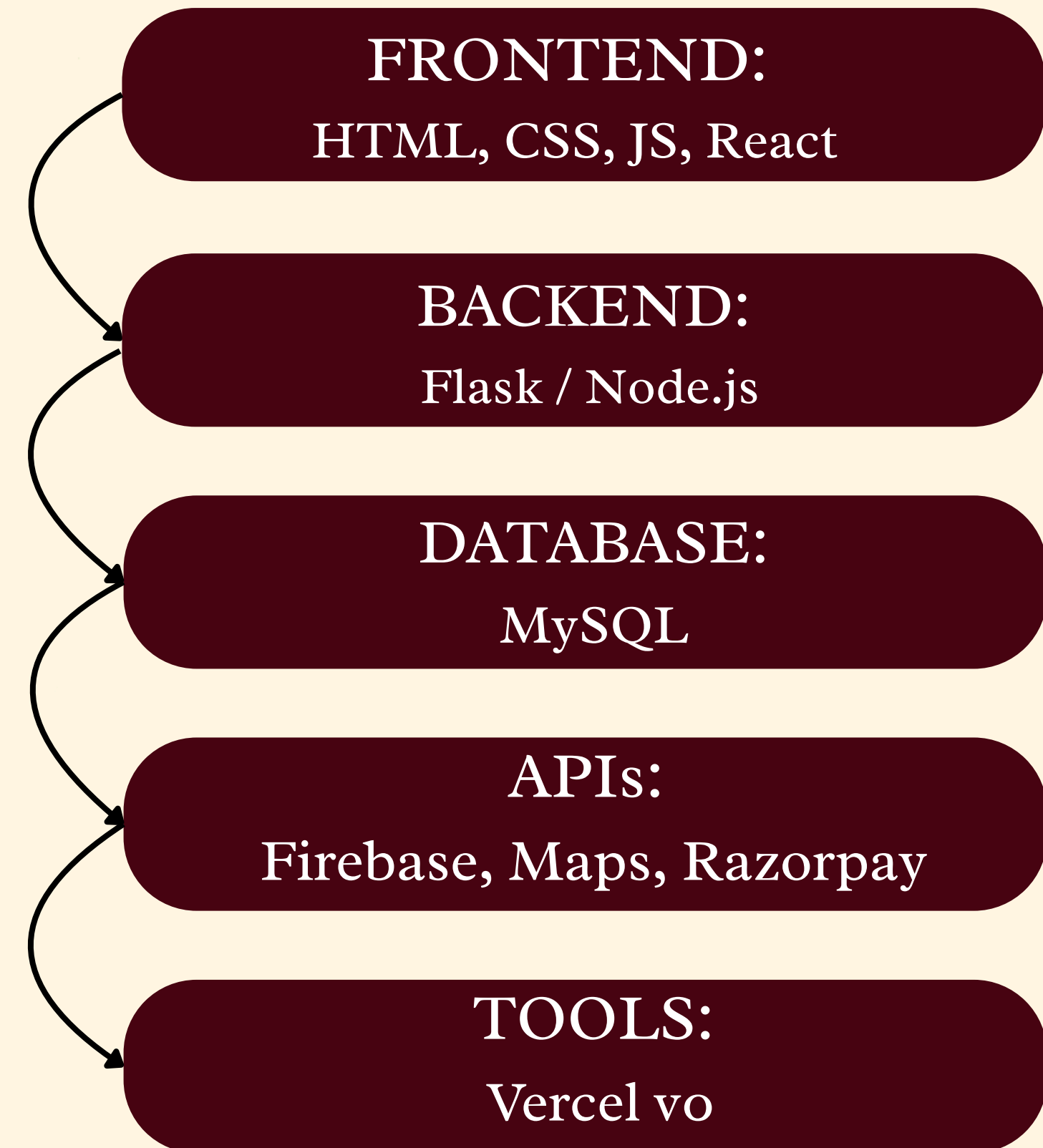
OUR SOLUTION

We propose a Smart LPG cylinder Management System that ensures transparency, fairness, and efficiency

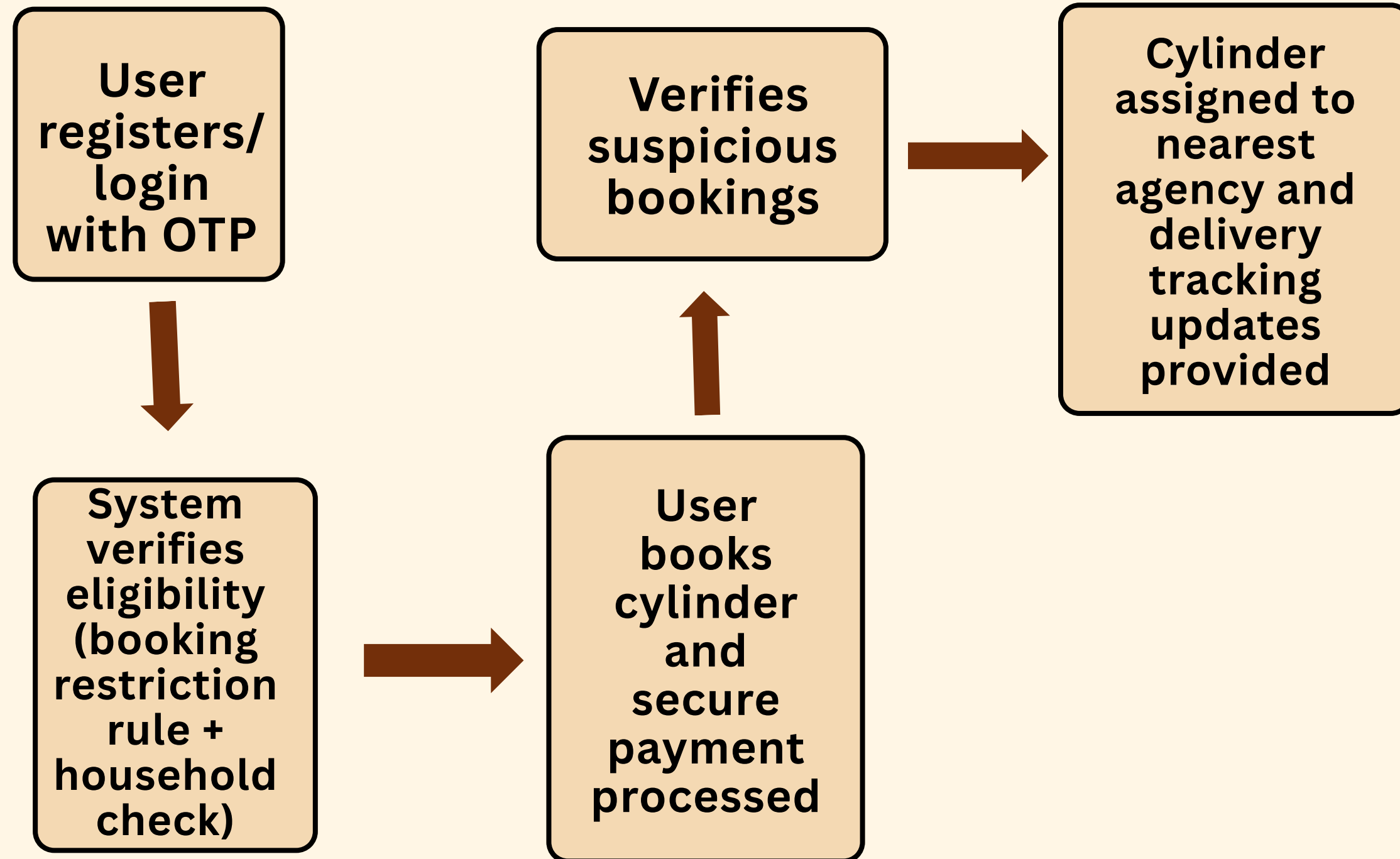
Key features:

- Household unit-based verification (multi-storey support)
- Smart booking eligibility based on family size
- Real-time stock visibility with dashboard
- Map-based LPG distribution & delivery tracking
- Secure booking(using OTP) & government pricing
- Fraud detection & admin verification system
- Collaboration across multiple LPG agencies
- Complaint, rating & monitoring system

TECHSTACK



USER FLOW

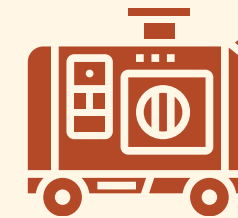


USP AND BUSINESS SUSTAINABILITY



USP

- Household **unit-level verification** (not just address-based)
- **Adaptive booking system** based on family size & real usage
- **End-to-end transparency** (stock + map + live tracking)
- Intelligent **fraud detection** with admin review (not blind blocking)
- Unified platform connecting **users, agencies & admin** systems

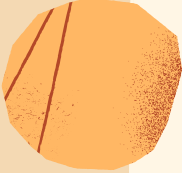


BUSINESS SUSTAINABILITY

- Can operate as a **SaaS platform** for LPG agencies & distributors
- **Government collaboration** for large-scale deployment
- **Low-cost cloud infrastructure**
- Revenue streams:
 - **Commission** per booking
 - Agency **subscription** model
 - **Data insights** for policy & optimization
- Reduces **losses from black marketing**

“TrackMyGas is not just a solution — it is a scalable digital infrastructure for fair resource distribution.”

CHALLENGES FACED DURING DEVELOPMENT



HOUSEHOLD-LEVEL VALIDATION

- Handling multi-storey buildings with multiple independent units
- Preventing misuse without blocking genuine households

SMART ELIGIBILITY DESIGN

- Implementing family-size adaptive booking rules
- Balancing fairness with real-world flexibility

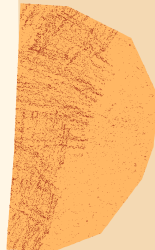
TRANSPARENCY SYSTEM IMPLEMENTATION

- Simulating real-time stock visibility + map-based distribution
- Ensuring consistency between stock, demand & delivery tracking

FRAUD DETECTION LOGIC

- Identifying duplicate accounts & abnormal booking patterns
- Designing flagging system instead of strict blocking

SYSTEM INTEGRATION

- Combining booking, tracking, admin & map features smoothly
 - Maintaining simple UX despite complex backend logic
- 





1

FUTURE SCOPE

- AI-based demand prediction & smart allocation
- Integration with government LPG databases
- Enhanced map system with real-time delivery optimization

2

SCALABILITY

- Expandable to national-level LPG distribution system
- Extendable to other public systems:
- Ration
- Water
- Electricity

3

FEASIBILITY

- Built using existing scalable web technologies
- Can be deployed as a district-level pilot
- Modular system → easy integration with agencies

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IMPACT

- Ensures fair, need-based LPG distribution
- Reduces black marketing & artificial shortages
- Builds trust through transparency (stock + tracking + map)



THANK YOU!

TrackMyGas~ “Not just delivering gas, but delivering fairness.”

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